UM Hackathon 2026

# FREE RIDERS

## *AI-Powered Food Waste Redistribution Platform*


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# INTRODUCTION

**“Connecting surplus food to those who need it — in real time”**

While restaurants throw away surplus food every day, many people go hungry just a few kilometers away. This isn’t because there isn’t enough food , but because there is no efficient way to connect those who have excess with those who need it, in time. Today, food donation is often manual, unstructured, and slow. By the time coordination happens, the food is no longer usable.

By leveraging **AI agents**, we can automate this entire process , starting with understanding natural language input from restaurants, structuring the data, and intelligently matching it to the most suitable NGOs in real time. With the power of AI, we can bridge this gap, ensuring surplus food reaches the right people, at the right time.

# PROBLEM STATEMENT

Food is wasted daily while many still go hungry. The problem isn't supply, it's coordination.

# SOLUTION

An AI agent that intelligently connects surplus food from restaurants to the right NGOs in real time.

# KEY FEATURES OF OUR SYSTEM

- **Natural Language Input**

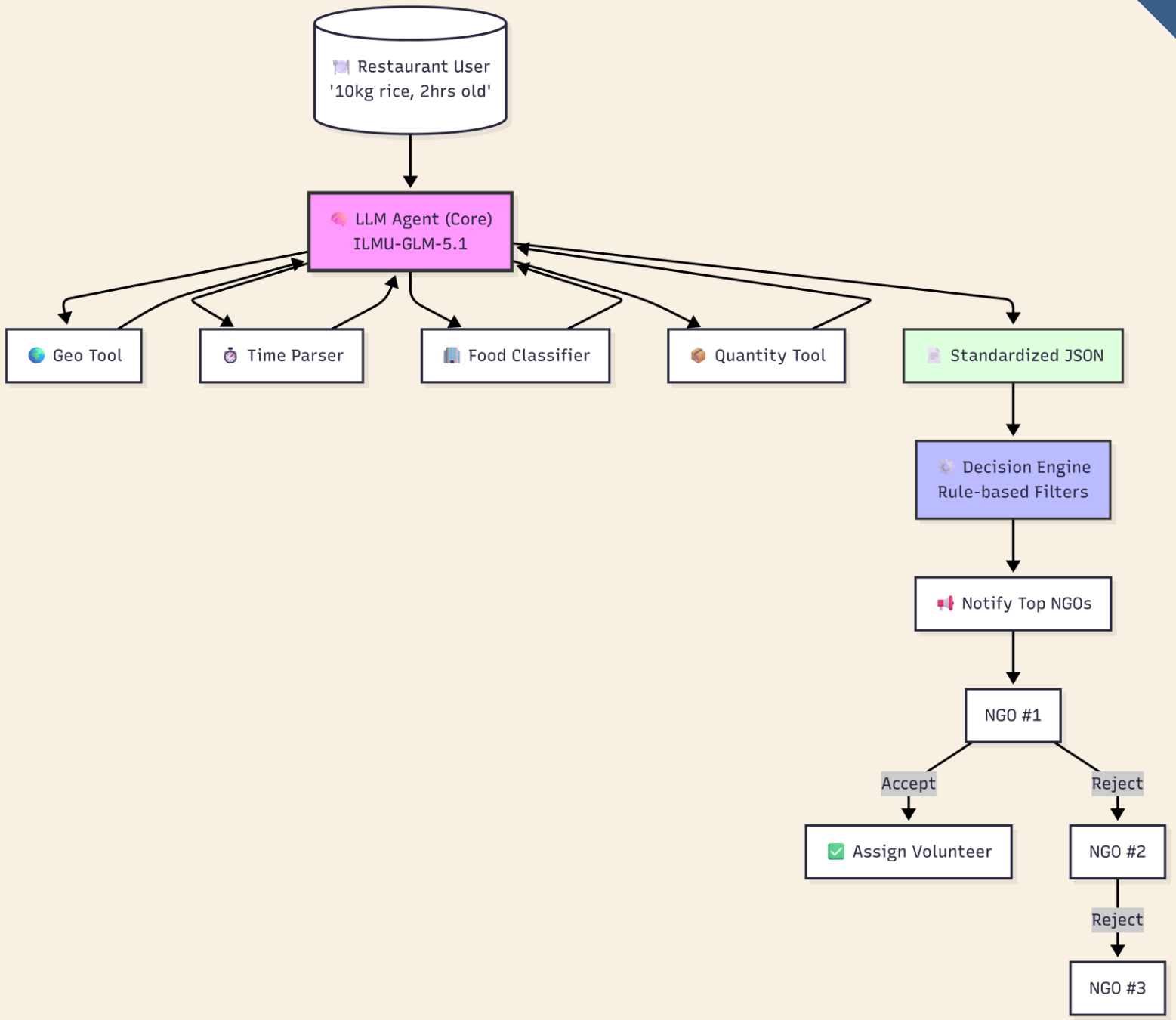
  Restaurants can easily describe surplus food in plain text. No complex forms or manual data entry required.
- **AI-Powered Data Extraction**

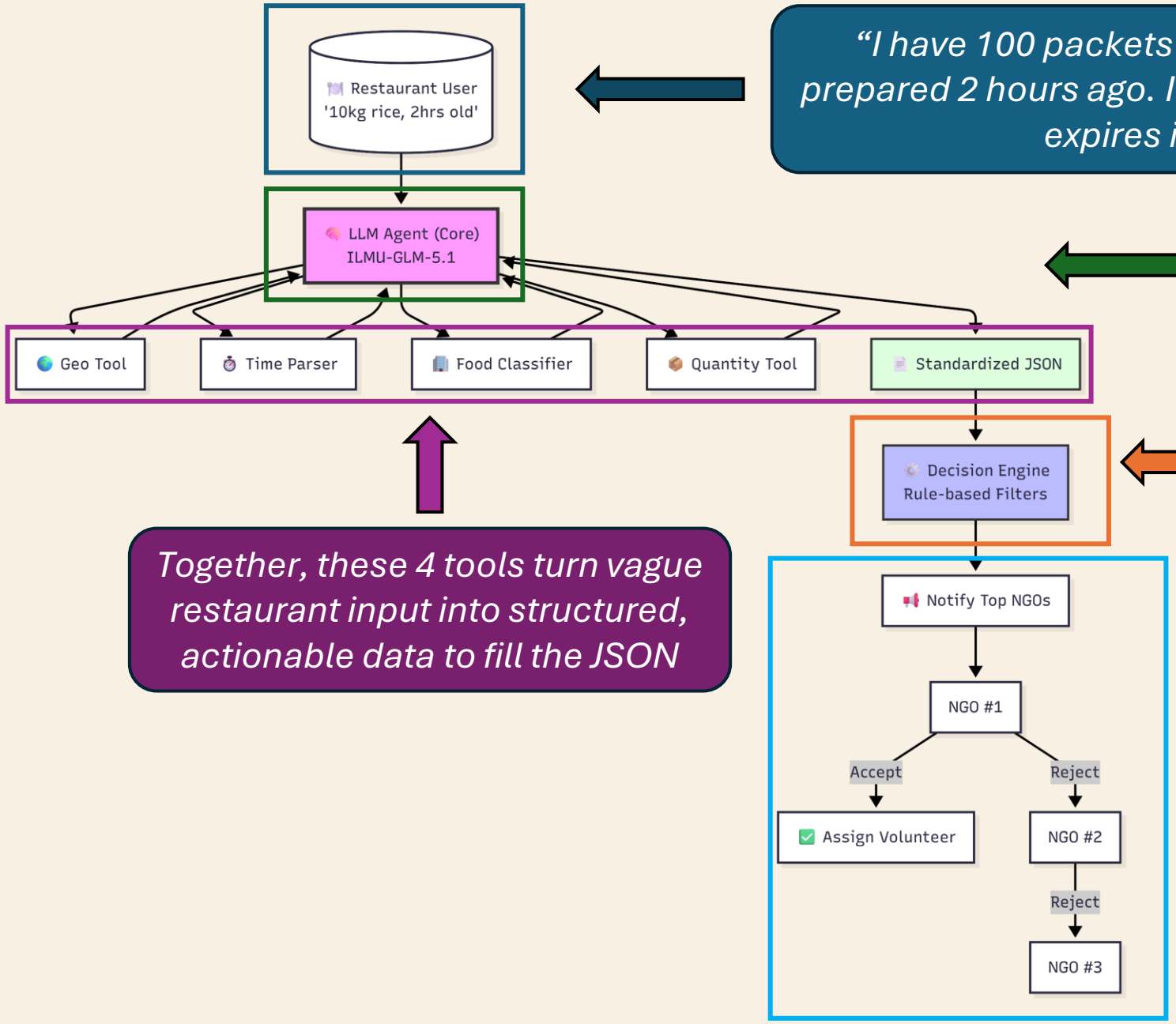
  An AI agent processes unstructured input and converts it into structured JSON (e.g., food type, quantity, expiry time).
- **Intelligent NGO Matching**

  A decision engine evaluates factors such as distance, urgency, and NGO capacity to identify the most suitable recipient.
- **Real-Time Notification System**

  Automatically alerts selected NGOs instantly, enabling faster response and reducing the risk of food expiry.

# AI AGENT ARCHITECTURE DIAGRAM






*"I have 100 packets of leftover packed rice prepared 2 hours ago. I want to donate it before it expires in 2 hours."*

*LLM fills in the JSON first, then calls 4 tools instantly if needed*

*The Decision Engine takes the completed JSON and runs it through 3 filters , such as distance, capacity, and expiry to then ranks the top NGOs*

*Together, these 4 tools turn vague restaurant input into structured, actionable data to fill the JSON*

Automatically notifies the most suitable NGO in real time

# DIAGRAM EXPLANATION

Our system starts with the **restaurant**, where users simply describe their surplus food in natural language – for example, the type of food, quantity, and expiry time.

This input is handled by our **AI agent**, powered by a large language model, which processes the unstructured text and converts it into a structured JSON format with the required fields.

If needed, the AI agent can also call external tools, such as location services or validation functions to enrich or verify the data. Once structured, the data is passed into our **decision-making engine**, which evaluates key factors such as distance, urgency, and NGO capacity.

Based on this, the system identifies the most suitable NGO and sends a **real-time notification** for food collection.

# WALKTHROUGH OF THE SYSTEM

# IMPACT

- **Reduce Food Waste**

  Enables timely redistribution of surplus food before it expires, minimizing unnecessary waste.

- **Support Underserved Communities**

  Helps NGOs deliver more consistent and reliable food aid to those in need.

- **Faster Response Time**

  Automates the donation process, reducing delays from hours to minutes.

- **Efficient Resource Utilization**

  Ensures food is matched to the right NGO based on real needs and capacity.

# WHAT'S NEXT

**"Turning surplus food into timely support, powered by AI."**

- **Autonomous Coordination with AI Agents**
  Enable AI agents to negotiate, confirm, and schedule pickups automatically between restaurants and NGOs.

- **Smart Logistics & Route Optimization**
  Leverage AI to plan optimal pickup routes, reducing delays and maximizing collection efficiency.

- **Predictive Surplus Intelligence**
  Use AI to forecast food surplus patterns and proactively match NGOs before waste occurs.

- **Continuous Learning & Feedback Loop**
  Implement a learning system where the AI agent improves over time using past donation data, NGO responses, and successful matches to enhance future decision-making accuracy.

# THANK YOU